

Teaching Phonological Theory using Automatic Speech Recognition

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Roadmap

1. Motivation
2. Our approach
3. Procedure
4. Analysis
5. Conclusion

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Phonological theory

Teaching phonology and phonological theory is challenging

- Exemplary syllabus for introductory course:

Phonetic preliminaries
Phonemes
Features
Phonological alternations
Syllables
Stress and Intonation

Table 1: Basic schedule for phonology intro courses

- Contemporary phonological theories may be covered if time permits
- Emphasis on abstract sound patterns and problem solving
- How to best teach abstract concepts? e.g. programming? (Tucker et al., 2024)

Phonological theory

- Phonology requires several abstract theoretical concepts right from the start (phonemes, allophones, representations, rules...)
- Approach involving aspects of sound structure and problem sets has not changed much in nearly 40 years, e.g. Gussenhoven & Jacobs (2017), Hayes (2009), Katamba (1989), and Zsiga (2024)
- If several phonological theories are covered: [confusing!](#)
- This approach is timeless and does a very good job in introducing the field's central ideas!
- **This approach, however, does not address the students' interests!**

Students' perspective

Students often question the use of learning phonological theory

- Too abstract, thus too disconnected from real language
- Too many theoretical approaches to cover all
- Less engagement → loss of interest

Students' perspective

How to best teach abstract subjects such as phonology?

- Skill-based teaching (Zuraw et al., 2019)
- Project-based teaching (Kakolu Ramarao et al., 2024)
- **Increasing student engagement → better learning**

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Our approach I

- Our approach made use of phonetic data and Automatic Speech Recognition (ASR)
 - address students' growing interest in language technology and computational linguistics
 - ASR for processing spoken language both as a tool as well as phonetic research (Emara & Shaker, 2024)
- We taught a 2 hour/week course for 13 weeks in summer 2025
- Elective course on Applied Phonology for German students
- Undergraduate students in General Linguistics + English & American Studies
- Central idea: Have students record their **own non-native English speech** and **evaluate output of ASR** transcriptions using **phonological rules and features**
- 20 registered students; 14 successfully finished the course

Our approach II

- Combined content from three fields
 1. Introduction to Acoustic Phonetics
 2. Introduction to Automatic Speech Recognition
 3. Introduction to Phonology

Central aspects

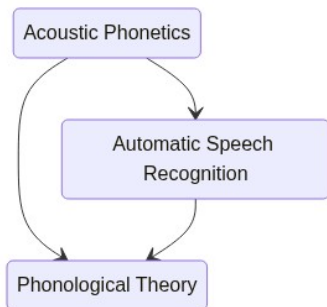


Figure 1: Combined approach

Our syllabus

#	Topic
01	Introduction & Logistics
-	HOLIDAY (no class)
02	Guided lab tour
03	Recordings I
04	Recordings II
05	Huggingface & Transcription
06	Introduction to Praat
-	STUDY WEEK (no class)
07	Phonology I: Recap
08	Phonology II: Processes
09	Phonology III: Features
10	Analysis I: Quantitative
11	Analysis II: Qualitative
12	Analysis III: Comparison
13	Wrap Up & Term paper

Material

- Sentences from **Artie bias corpus** (Meyer et al., 2020)
 - Natural sentences used for ASR benchmarking
- 70 sentences per speaker
- Examples:
 - *No, we don't laminate our business cards.*
 - *Love is a fluid concept, and I'm not sure if androids get that.*
 - *The python wasn't a snake, it was a programming language.*
 - *Two-wave sources are perfectly coherent if they have a constant phase difference and the same frequency.*

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Procedure I

- Students were grouped based on their native language(s)
- Students recorded each other in our phonetics lab using *Audacity* (Team, 2025)
- Different ASR models were assigned to different students
- We selected models that differ in size, architecture, and how faithful the transcriptions were to the signal

Model	Labelled training data	Parameter size
Whisper-small (Radford et al., 2023)	680k hours	244 million
Whisper-large (Radford et al., 2023)	680k hours	1550 million
Crisper-Whisper (Wagner, Thallinger, & Zusag, 2024)	680k hours	finetuned whisper model
Parakeet (Sekoyan et al., 2025)	670k hours	600 million

Procedure II

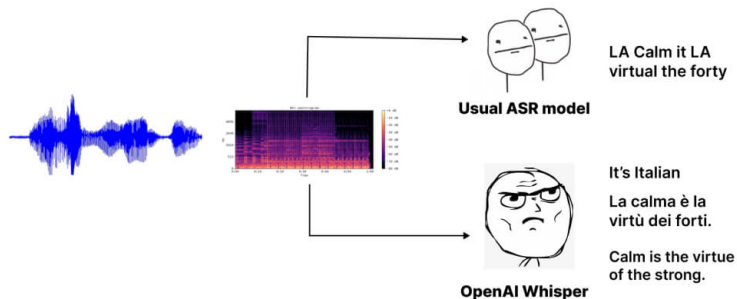


Figure 2: <https://www.superannotate.com/blog/openai-whisper-automatic-speech-recognition-system>

Procedure III

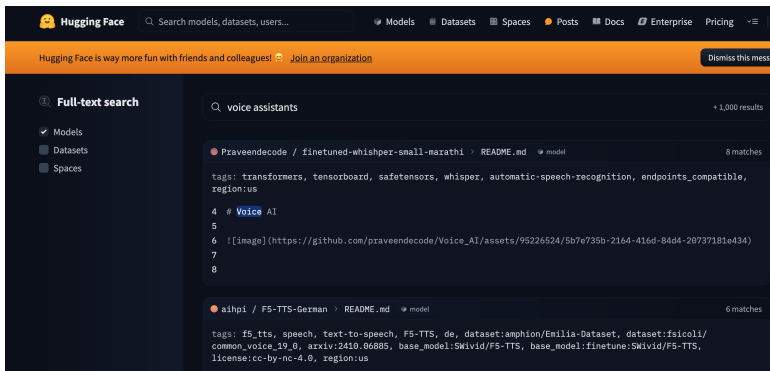


Figure 3: Testing Huggingface models

Procedure IV

- Students created annotated TextGrids based on ASR transcripts by using *Praat* (Boersma & Weenink, 2025)
- TextGrids with four interval tiers
 - Target-sentence
 - ASR-sentence
 - Target-word
 - ASR-word
- For each sentence-aligned tier, one contained the target productions, the other corresponding ASR transcripts
- Word-aligned tiers were the same, but only words with a mismatch between target and transcription were annotated

Procedure V

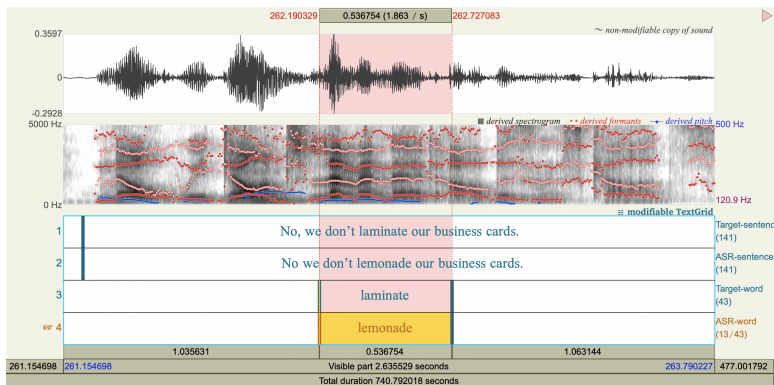


Figure 4: Acoustic analysis and transcription in Praat (Boersma & Weenink, 2025)

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Phonological analysis I

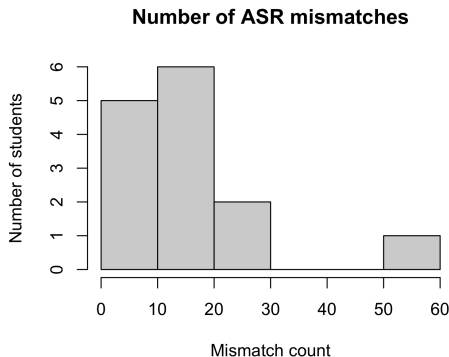


Figure 5: Number of word errors varied depending on ASR model and speaker data

- Annotations for 14 students
- Mean mismatches = 16.3 (SD $\pm$ 13.8)

Phonological analysis II

- Students proposed phonological processes that they analysed in analogy to rule-based phonological changes
- Phonological rules were written both as segments and features
- Rules were practiced with processes for Standard German using the *Pheatures* software (van Vugt, Hayes, & Zuraw, 2012)
- This facilitates exploration to see if differences in segmental inventory and processes will affect ASR transcriptions
- Some transcription errors related to phonological transfer, but some were unrelated to phonology

Phonological analysis III

Select Features

value

Feature

-

sonorant

value

Feature

+

voice

value

Transform Features

value

Feature

-

voice

value

Feature List

☐	Phoneme	syllabic ↓	stress	long	consonantal	sonorant	continuant	delayed release	approximant	tap	trill	nasal	voice	spread gl	const
☐	b → p	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	d → t	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	g → k	-	-	-	+	-	-	-	-	-	-	-	-	-	-
☐	v → f	-	-	-	+	-	+	+	-	-	-	-	-	-	-
☐	z → s	-	-	-	+	-	+	+	-	-	-	-	-	-	-
☐	ʁ → χ	-	-	-	+	-	+	+	-	-	-	-	-	-	-

Figure 6: Using *Pheatures* (van Vugt, Hayes, & Zuraw, 2012)

Online application (credits Taesung Hwang):
<https://pheatures.slam.phil.hhu.de/>

Phonological analysis IV

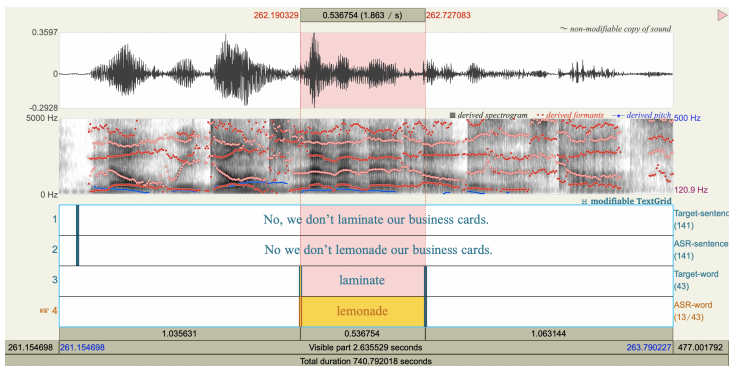


Figure 7: Acoustic analysis and transcription in Praat (Boersma & Weenink, 2025)

Target pronunciation	ASR transcription	Process (Vowel raising)
/æ/	[ɛ]	/æ/ → [ɛ]
[-back, +low, -high, -ATR] →	[-back, -low, -high, -ATR] →	[-back, +low, -high, -ATR] → [-low]

Quantitative analysis

For model comparison, quantitative analyses using standard ASR evaluation metrics were also conducted

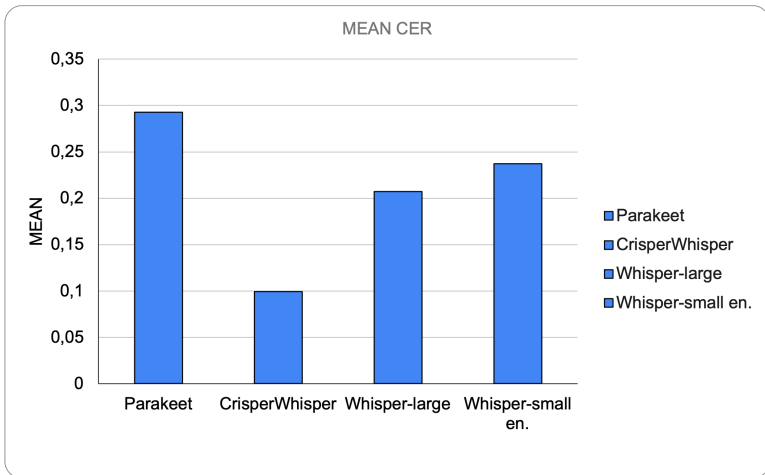


Figure 8: Results Character Error Rate (CER) by Model

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Learning outcomes I

Conceptual topics:

- Technical aspects of recording speech in a phonetics laboratory
- The basics of acoustic phonetics
- The basics of automatic speech recognition
- An introduction to the acoustic-phonetic software *Praat*
- Conceptual differences between phonetics and phonology in the context of German and English
- Comparison of ASR model performance using standard ASR evaluation metrics

Learning outcomes II

Technical skills:

- Conducting speech recordings in a phonetics laboratory
- Using the acoustic-phonetic analysis software *Praat*
- Working with ASR models on the HuggingFace platform (Parakeet, Whisper-small, Whisper-large, CrisperWhisper)

Student feedback I

Qualitative measure:

- Anonymous student feedback: hands-on aspects and course structure contributed to learning success
- One student experienced the course as too theory-heavy, with other students finding the work sheets complicated
- Several students faced difficulties in analyzing sound changes with rules and features
- Fine tuning is needed though:
 - Students with stronger English accent had more mismatches between target and ASR transcription
 - Work around this by:
 - Using simpler models that produce more transcription errors
 - Using spontaneous speech instead of read speech

Student feedback II

Quantitative measures:

Ihre Gesamtbewertung

Alles in allem: Wie bewerten Sie diese Veranstaltung bisher?

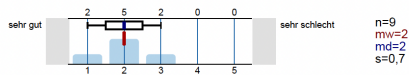


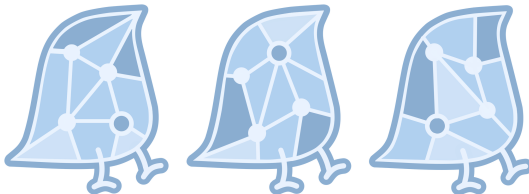
Figure 9: Overall student ratings (from 1-5, with 1 as best grade)

Takeaway

- We taught a two hour/week phonological theory class to undergraduate Linguistics and English students which made use of Automatic Speech Recognition (ASR)
- This project-based approach to learning about phonology held promising results
- We advocate for thinking about novel, timely ways to teach phonology
 - Incorporating language technology such as ASR + phonetic data may help connect students with phonology
 - Connecting theoretical notions with real data → more engagement and better learning

Acknowledgement

- Special thanks to our undergraduate students that participated in this project!
- Slamlab (<https://slam.phil.hhu.de/>) for providing the lab facility



End

Thank you for your attention!



- Scan QR code to access slides

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Appendix

Appendix

Results course evaluation

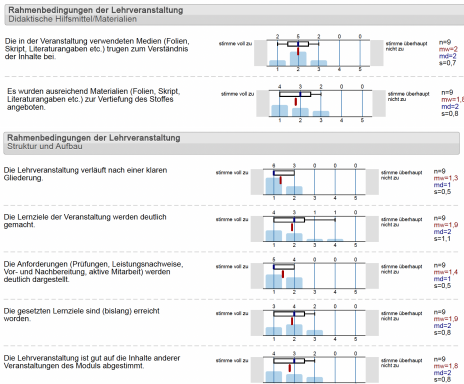


Figure 10: Course evaluation I

Results course evaluation

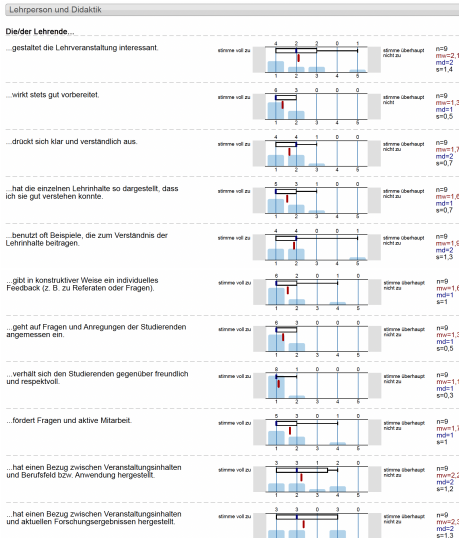


Figure 11: Course evaluation II

Results course evaluation

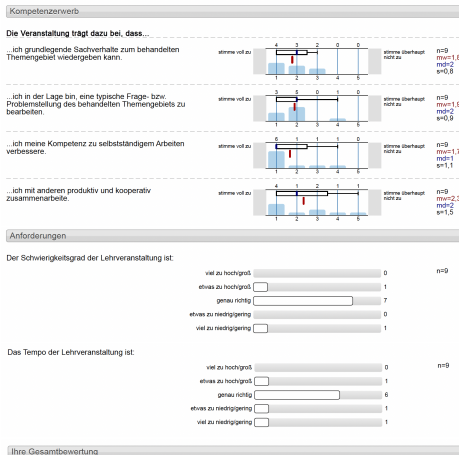


Figure 12: Course evaluation III

ASR metrics I

Character Error Rate (CER)

Calculates best alignment that minimises the number of substitutions, insertions and deletions necessary to map the reference transcript to the hypothesis transcript.

$$CER = \frac{S + D + I}{N} \quad (1)$$

where,

S = number of substitutions

D = number of deletions

I = number of insertions

N = total number of characters in reference

ASR metrics II

Word Error Rate (CER)

Calculates best alignment that minimises the number of substitutions, insertions and deletions necessary to map the reference transcript to the hypothesis transcript.

$$WER = \frac{S + D + I}{N} \quad (2)$$

where,

S = number of substitutions

D = number of deletions

I = number of insertions

N = total number of words in reference

Quantitative analysis ASR I

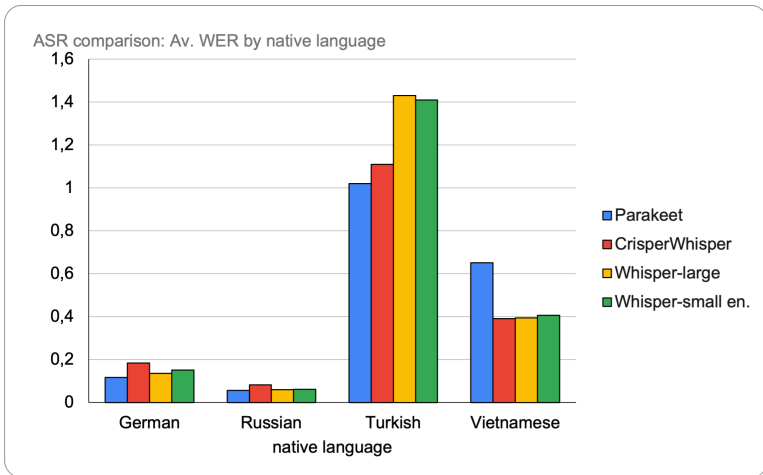


Figure 13: Results Word Error Rate (WER) – Language

Quantitative analysis ASR II

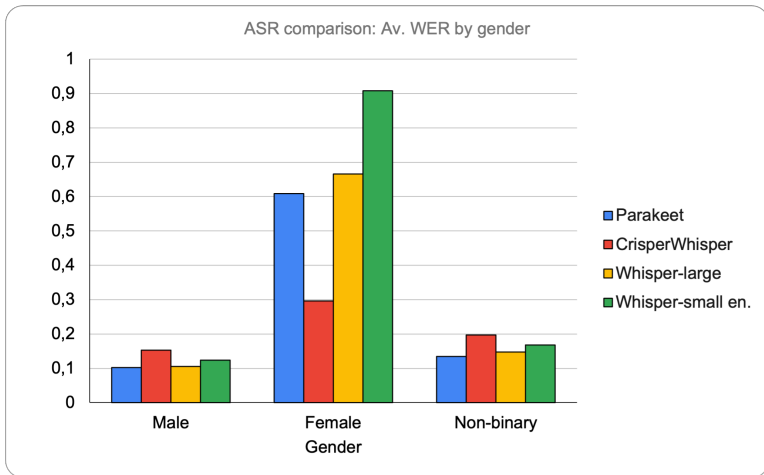


Figure 14: Results Word Error Rate (WER) – Gender

Quantitative analysis ASR III

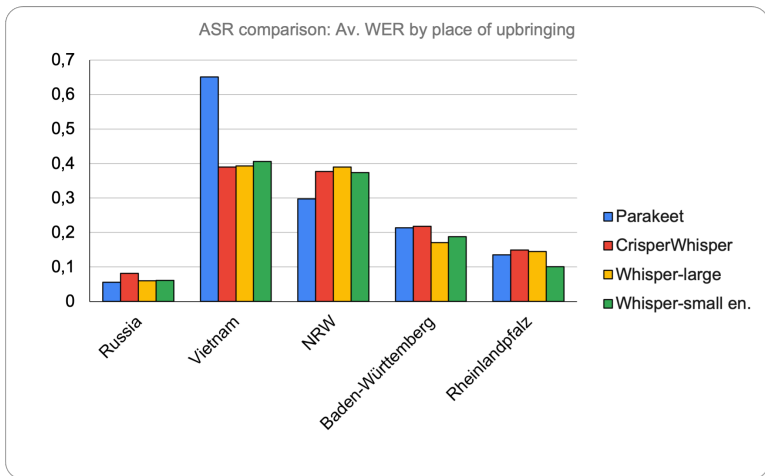


Figure 15: Results Word Error Rate (WER) – Upbringing